

Conformal Inference Assignment 1

(Deadline: September 29)

Wen, Zehai

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Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 1

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$ and a given model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}$. Construct a score function $s(x, y)$ that would yield intervals of the form $\mathcal{C}(x) = [\hat{f}(x) - \gamma, \hat{f}(x) + 2\gamma]$ for some $\gamma > 0$.

Solution Put

$$s(x, y) = \begin{cases} 2|\hat{f}(x) - y| & y < \hat{f}(x) \\ |\hat{f}(x) - y| & y \geq \hat{f}(x). \end{cases}$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$, we will have the desired interval with $\gamma = \frac{\hat{q}}{2}$. ///

Exercise 2

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}^d$ and a given model $\hat{f} : \mathcal{X} \rightarrow \mathbb{R}^d$ whose coordinates are $\hat{f}(x) = (\hat{f}_1(x), \dots, \hat{f}_d(x))$. Construct a score function $s(x, y)$ that would yield intervals of the form

$$\mathcal{C}(x) = \prod_{j=1}^d [\hat{f}_j(x) - c, \hat{f}_j(x) + c]$$

for a $c > 0$.

Solution Put

$$s(x, y) = \max_{j=1, \dots, d} |\hat{f}_j(x) - y_j|.$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\}$, we will have the desired interval with $c = \hat{q}$. ///

Exercise 3

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$, two given models $\hat{f}_i : \mathcal{X} \rightarrow \mathbb{R}$ for $i = 1, 2$, and the score function

$$s(x, y) = \min_{i=1,2} |y - \hat{f}_i(x)|.$$

Derive an explicit expression for $\mathcal{C}(x)$ in terms of $\hat{f}_1(x)$, $\hat{f}_2(x)$, and the quantile $\hat{q}$ of the scores on the validation set. What is the shape of the prediction when $\hat{q}$ is small or when it is large? Briefly explain why this might be attractive if the conditional distribution of Y given $X = x$ is bimodal.

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

Exercise 4

Fix $n > 2$. Construct a random vector $(X_1, \dots, X_n)$ defined on some probability space such that (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \dots, X_n)$ is not exchangeable.

Solution Let (X, Y) be a jointly normal random vector. It is exchangeable if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\text{Var}(X) = \text{Var}(Y)$. With this in mind, let $(X_1, \dots, X_n)$ be a jointly normal random vector such that $\mathbb{E}[X_i] = \mathbb{E}[X_j]$ and $\text{Var}(X_i) = \text{Var}(X_j)$ for all $i \neq j$ but $\text{Cov}(X_1, X_2) \neq \text{Cov}(X_1, X_3)$. Then (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \dots, X_n)$ is not exchangeable. For a concrete example, say $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) = 1$ for all i , $\text{Cov}(X_1, X_2) = -1$, and all other covariances are zero. ////

Exercise 5

Fix $n \geq 1$, $\tau \in (0, 1)$. For any vector $v \in \mathbb{R}^n$, define $\text{Quantile}(v; \tau) = v_{(\lceil \tau n \rceil)}$, where $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(n)}$ are the order statistics of v . Under what condition is it true that, for all $v \in \mathbb{R}^n$, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$?

Solution The order statistics of $-v$ are given by $(-v)_{(i)} = -v_{(n-i+1)}$ for $i = 1, \dots, n$. Therefore, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$ holds for a vector $v \in \mathbb{R}^n$ if and only if

$$v_{(\lceil (1-\tau)n \rceil)} = -((-v)_{(\lceil \tau n \rceil)}) = -(-v_{(n-\lceil \tau n \rceil+1)}) = v_{(n-\lceil \tau n \rceil+1)}.$$

To make this work for all $v \in \mathbb{R}^n$, the condition we must pose is that

$$\lceil (1 - \tau)n \rceil = n - \lceil \tau n \rceil + 1.$$

This can happen for instance when $\tau = 1/2$ and n is odd. ////

Exercise 6

Prove all parts of fact 2.12

Solution NEED WORK ////

Exercise 7

exercise 3.1

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

Exercise 8

exercise 3.6

Solution By definition, it is immediately clear that

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Exercise 9

exercise 3.7

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leq \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x , that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////